Alex Caruana

Conservation Scientist

I am a conservation scientist with a strong passion for biodiversity conservation and environmental planning. I have strong data analysis skills (R and QGIS) and a focus on doing rigorous research with applied relevance.

alexcaruana97@gmail.com | Oxford, United Kingdom | alexcaruana.com

EDUCATION

Sep. 2024 - Ongoing
D.Phil. Conservation Biology
University of Oxford (United Kingdom)
Thesis Title: "Advancing counterfactual thinking within conservation science."
Thesis Supervisor: Professor Joseph Bull, Secondary co-supervisor: Professor Julia P. G. Jones

Sep. 2021 - Sep. 2023
M.Sc. Sustainable Forest and Nature Management specialised in Conservation Biology and Land Management
University of Copenhagen (Denmark) & Bangor University (United Kingdom)
Degree Classification: Distinction
Thesis Title: "Lessons Lost – Lack of post-project monitoring hinders evidence-based conservation".
Thesis Supervisor: Professor Julia P. G. Jones (grade: 87%)

Oct. 2018 - Jun. 2021
B.Sc. Environmental Engineering
Malta's College of Arts, Science and Technology (Malta)
Degree Classification: First Class Honours
Awards: *Outstanding Performance* Award
Thesis Title: "Assessing the effectiveness of seeding in large-scale ecological restoration of *Periplocion angustifoliae* maquis".
Thesis Supervisor: Dr. Eman J. Calleja

PUBLICATIONS

Peer-review Papers

1. Shackelford, N., [and 76 others, including **Caruana, A.**] et. al. 2021. Drivers of seedling establishment success in dryland restoration efforts*. Nature Ecology and Evolution.* https://doi.org/10.1002/2688-8319.12325 [**Published**]

2. **Caruana, A.**, Camilleri, B., Farrugia, L., Jones, JPG. 2024. Mechanical excavation of wetland habitat failed to eradicate invasive American red swamp crayfish (*Procambarus clarkii*) in Malta. *Ecological Solutions and Evidence*. https://doi.org/10.1002/2688-8319.12325 [**Published**]

3. **Caruana, A.**, Muir, M., White, T., Jones, J. P. G. Lessons lost: Lack of requirements for post-project evaluation and reporting is hindering evidence-based conservation, Conservation Science and Practise. https://doi.org/10.1111/csp2.13260 [**Published**].

Other Reports & Datasets

1. **Caruana, A.** and Farrugia, L. (2023) 'Assessment of Red Swamp Crayfish presence, distribution and abundance within the Fiddien valley system'. *Applied Ecological Resources*. https://www.britishecologicalsociety.org/applied-ecology-resources/document/20240157815/ [**Published**]

CONFERENCES

1. **Caruana, A.**, Jones, J.P.G., Wauchope, H., Christie, A.P., Bull, J. The core assumptions of causal inference. Oral presentation delivered at the Society for Conservation Biology's 32nd International Congress for Conservation Biology (ICCB 2025), June 2025.
2. **Caruana, A**. and Calleja, E. Assessing the effectiveness of seeding as a cheap large-scale restoration tool in a semi-arid Mediterranean habitat. Oral presentation delivered (online) at the *World Conference on Ecological Restoration*, 24th June 2021.

WORK EXPERIENCE

Sep. 2020 - Sep. 2021 **Environmental Specialist (Auditor)**, MPS Limited (Malta)
Executed in-depth water audits within households, agricultural farms, and governmental buildings to improve water-use efficiency. Wrote technical reports and documentation.

Jan. 2019 - Sep. 2020 **Environmental Intern (Research and Monitoring)**, ADI Associates and En-sure Limited (Malta)
Provided GIS support, producing maps for Environmental Impact Assessments. Conducted site surveys and ecological assessments for over 100 water bodies throughout the Maltese Isles for Malta's MedIsWet Project. Analyzed data and wrote reports.

Jun. 2017 - Sep. 2018 **Environmental Research Assistant**, Ministry for Environmental Sustainability, the Environment and Climate Change (Malta)
Provided GIS support, producing maps for Environmental Impact Assessments. Conducted site surveys and ecological assessments for over 100 water bodies throughout the Maltese Isles for Malta's MedIsWet Project. Analyzed data and wrote reports.

SCHOLARSHIPS AND FUNDING

Oxford's Clarendon Scholarship (2024 – 2027)	€ 67,490
Malta's Endeavour Scholarship Scheme (2021 – 2023)	€ 16,791
World Conference on Ecological Restoration Ticket funded via MCAST (2021)	€ 189
Bangor University's School of Natural Science Master Thesis Fund	€ 288
Total Funding	**€ 84,758**

OTHER

Technical skills:
- Data Analysis via R Programming (https://github.com/AlexCaruana)
- Geographical Information Systems (QGIS & ArcGIS)
- Design Programs (Adobe Photoshop, Illustrator & InDesign)
- General Software (Microsoft Word, Excel & PowerPoint)

Licenses and Certificates:
- Driving License (B)
- Online Certificate in Mapping, Spatial Data and GIS (University of Oxford)
- Global and Regional Assessor Certificate (IUCN)
- Data Science: Foundation using R Certificate (John Hopkins University via Coursera.com)

Membership of Professional Associations:
- British Ecological Society (since June 2023)
- Society for Conservation Biology (since September 2019)
- Society for Ecological Restoration (since September 2019).

Languages: English (Native), Maltese (Native)

VOLUNTEER EXPERIENCE

Dec. 2022 - Jan. 2023 **Voluntary Research Assistant**, University of Edinburgh (Scotland)
Contributing to a systematic review of community-based protected area law enforcement led by William Sharkey and supervised by Dr. Aidan Keane and Prof. Dame E.J. Milner-Gulland.

Oct. 2020 - May 2021 **Voluntary Research Assistant**, Shark-Lab (Malta)

Gathering data on shark and ray populations around the Maltese Isles by surveying the industrial fish market. Contributing to reintroduction efforts by harvesting any remaining egg cases from deceased sharks and transporting them to the National Aquarium located in Qawra for incubation.

Aug. 2019 - Aug. 2021 **Ecological Restoration Volunteer**, Nature Trust (Malta)

Organized and aided in ecological restoration activities within Il-Magħluq ta' Marsaxlokk. Activities included regenerating the sand-dune barriers with bio-engineered structures, sowing endemic species in deteriorated areas, and eradicating invasive alien species.